Reg. No.:

Name :

VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Continuous Assessment Test II – March 2026

Programme	: B.Tech (CSE & its Specialised Branches)	Semester	:	WS 2025-26	
Course	: Embedded Systems	Code	:	BCSE305L	
		Class Nbr	:	CH2025260502873	
Faculty	: Dr. Sritama Roy	Slot	:	G2+TG2	
Time	: 90 Minutes	Max. Marks	:	50	

Answer ALL the questions

Q.No.	Sub. Sec.	Questions	Marks	CO's	BT Level
1.		The characters a through h have the following frequencies. a : 2, b : 3, c : 5, d : 7, e : 11, f : 13, g : 17, h : 19 A Huffman code has been generated for these characters. a) During transmission of the encode the following characters into the Huffman code. **(5 Marks)** adhhgehdabchgfe b) Using this Huffman code, decode the following bit sequence into the original sequence of characters. **(5 Marks)** 111011110011101000	10	3	K3
2.		A space rover system is controlled by a central control unit and includes components such as navigation sensors, cameras, and a ground control application. The system enables scientists to remotely control the rover, monitor environmental conditions, and capture images or data from the planetary surface. a) Draw a model that accurately represents the system structure of the space rover. Your model should illustrate the relationships between the classes Rover, Sensor, ControlUnit, and GroundControlApp, and include functionalities such as movement control, data collection, and obstacle detection. **(7 marks)** b) Model the space rover system using places, arcs, and tokens to represent how the space rover operates during exploration, capturing images, and data transmission with neat diagram. **(8 marks)**	15	3	K3

3.		An embedded system used in a wearable health monitoring device continuously reads sensor data and processes it to display results. The following code segment is used in the system.
		
		`int sum = 0;`
		`for(int i = 0; i < 1000; i++)`
		`{`
		`  sum = sum + sensorData[i] * 2;`
		`}`
		
		Identify the possible inefficiencies in the above code when used in a resource
			5	3	K2

constrained embedded system and suggest an optimized version of the code to improve performance and explain why it is more efficient.

4. Determine whether an RTOS, GPOS, or EOS is necessary for the following real-time applications. If an RTOS is required, justify your choice and specify the type of RTOS most suitable for the application.

(i) Smart Thermostat
(ii) Wearable Fitness Tracker
(iii) Autonomous Drone Navigation System
(iv) Home Voice Assistant
(v) Automated Coffee Machine

5 2 K3

5. A smart home automation system executes several periodic tasks as listed below.

a) Assign priorities to each task. **(2 Marks)**
b) Verify the schedule using the necessary and sufficient schedulability test. **(3 Marks)**
c) Create a schedule using Gantt chart applying the Rate Monotonic Scheduling (RMS) and Earliest Deadline First algorithm. **(10 Marks)**

Task	**Computation Time (C)**	**Period (T in ms)**	**Priority**
Camera control	4	20	
Limb control	2	15	
Voice	1	5	
Gesture	3	25	
Expression control	2	10	

15 2 K3

Course Faculty

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